Cloud-Based Intelligent Document Forgery Detection and Verification System

DocuShield: A local, evidence-led prototype combining document forensics, OCR text extraction, cryptographic fingerprinting, and exact file verification.

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Academic Program:	Master of Technology in Computer Science & Engineering Specialization: *Data Science*
Course:	Cloud Computing
Project Category:	Software-Only Prototype (Hardware Requirement: None)
System Core:	FastAPI REST Engine • PyMuPDF • OpenCV • Scikit-learn • SQLite • Vite/Web Dashboard

Key Architectural Principle: *Integrity checks answer different questions; no single heuristic establishes authenticity. DocuShield strictly distinguishes between* ***exact cryptographic file identity*** *(SHA-256 byte-level MATCH/MISMATCH) and* ***document forensic indicators*** *(metadata completeness, Laplacian sharpness, edge density, and OCR linguistic patterns). The final output is an explainable audit report for human decision-makers.*

1. Executive Summary & Problem Context

Digital certificates, invoices, academic mark sheets, and identity documents are routinely transmitted in PDF and image formats. Modern raster editing and PDF manipulation tools make it trivial to alter numerical amounts, student names, issuer stamps, and dates without leaving obvious visual blemishes. Manual inspection of every incoming record is labor-intensive, error-prone, and unscalable.

Existing automated solutions suffer from three fundamental pitfalls: (1) **Scattered checks**, where visual inspection, OCR text extraction, and metadata validation occur in disjointed tools; (2) **Mutable metadata**, where missing or altered metadata is prematurely treated as absolute proof of tampering; and (3) **Ambiguous hash mismatches**, where a single byte change signals that files differ, but cannot reveal why or how the document was changed.

2. Project Objectives & Deliverables

#	Objective	Implementation Mechanism	Status
01	Accept & Validate Files	PDF/PNG/JPG magic header verification & 25MB boundary checking	**Implemented**
02	Extract Document Signals	PyMuPDF stream & metadata extraction; Pillow/OCR pipeline	**Implemented**
03	Cryptographic Identity	64-hex SHA-256 fingerprinting matched against trusted registry	**Implemented**
04	Image Forensics	Laplacian sharpness, Canny edge density, Error Level Analysis (ELA)	**Implemented**
05	Visual Alteration Highlighting	Pixel-level delta thresholding & bounding-box contour detection	**Implemented**
06	Audit & Report Generation	SQLite audit trail and on-demand downloadable PDF certification	**Implemented**

3. Scientific Literature Survey & State of the Art

Copy-Move Detection *(Christlein et al., IEEE TIFS)*	Isolates duplicated image regions and cloned patches.	Focuses purely on image plane; overlooks PDF stream and metadata.	Integrated as pixel alteration & diff contour bounding layer.
Cryptographic Hashing *(FIPS PUB 180-4 SHA-256)*	Deterministic 256-bit digest confirming exact byte equality.	Zero tolerance: a harmless re-save causes total mismatch.	Acts as the primary exact identity layer against registered masters.
Error Level Analysis (ELA) *(Krawetz, Hacker Factor)*	Exposes differing JPEG compression levels in composite images.	Can produce false positives on natural high-contrast edges.	Used as a secondary visual clue for human forensic reviewers.
NLP Pattern Screening *(Scikit-Learn TF-IDF)*	Detects lexical anomalies and tampering keyword distributions.	Text alone cannot guarantee authentic origin.	Scores document patterns to flag suspicious phrases for review.

4. Mathematical & Forensic Feature Extraction

To perform reproducible document assessment, DocuShield executes multi-layer mathematical analysis across cryptographic, spatial, and frequency representations:

A. Cryptographic SHA-256 Hashing

For any document bitstream M, SHA-256 processes 512-bit blocks through 64 rounds of logical functions (Ch, Maj, Σ0, Σ1) using message schedules to yield a 256-bit collision-resistant digest H(M). In DocuShield, matching H(M_upload) == H(M_registered) proves bitwise identity with probability 1 - 2^(-128).

B. Laplacian Variance Sharpness Metric

The discrete 2D Laplacian operator L(x, y) = ∇^2 I(x, y) approximates the second spatial derivative of the grayscale image I. The variance of the Laplacian response quantifies the high-frequency edge transition energy:
Sharpness Variance: σ^2 = (1 / (W · H)) · Σ (L(x, y) - μ_L)^2
Scanned documents with blurry text or recaptured photo copies exhibit σ^2 < 100, whereas digitally rendered vector text yields σ^2 > 1000.

C. Canny Edge Density

Using dual hysteresis thresholds (T_low = 100, T_high = 200) following Gaussian smoothing, edge pixels E(x, y) ∈ {0, 1} are identified. The Edge Density ρ_E = (Σ E(x, y)) / (W · H) measures layout complexity and reveals spliced high-density text inserts.

D. Error Level Analysis (ELA) Compression Discrepancy

Lossy JPEG compression computes 8x8 DCT transforms and quantizes coefficients. When an edited region is pasted into an existing JPEG and re-saved, the pasted section undergoes 1st-generation quantization while the background undergoes 2nd-generation quantization. DocuShield computes Δ_ELA(x, y) = |I_orig(x, y) - I_recomp(x, y)| at quality 90 to visually isolate tampered zones.

5. System Architecture & Component Design

Component Layer	Technology	Functional Responsibility
Presentation Layer	HTML5 / Tailwind / JavaScript	Interactive single-page dashboard, drag & drop workspace, diff visualizer
REST API Gateway	Python FastAPI / Uvicorn	Multi-part file ingest, CORS management, validation, async routing
Document Parser	PyMuPDF (fitz) + Pillow	PDF stream decoding, font/metadata inspection, multi-page extraction
Forensics Engine	OpenCV + NumPy + SciPy	Laplacian sharpness, edge density, ELA analysis, pixel diff bounding boxes
ML Screening Pipeline	Scikit-learn (TF-IDF + LR)	Textual pattern risk classification and linguistic anomaly scoring
Data Persistence	SQLite (docushield.db)	Relational storage of documents, trusted masters, and audit history
Audit Report Engine	ReportLab Flowable PDF	Automated generation of publication-grade verification certificates

6. REST API Specification & Endpoint Contracts

The FastAPI backend strictly implements the 9 core endpoints specified in Project Review 2 (Slide 20):

`GET`	`/api/health`	Service health, version, OpenCV/PyMuPDF runtime availability (200 OK)
`POST`	`/api/documents`	Uploads document, validates magic bytes, computes SHA-256, runs pipeline
`GET`	`/api/documents?search=`	Queries stored records filtered by filename, hash, or extracted text
`GET`	`/api/documents/{id}`	Retrieves full forensic metadata, ELA metrics, and OCR string
`POST`	`/api/originals`	Registers authoritative master hash with label, category, and timestamp
`GET`	`/api/originals`	Enumerates all registered master fingerprints
`POST`	`/api/documents/{id}/verify/{orig_id}`	Executes cryptographic MATCH/MISMATCH check & pixel diff
`GET`	`/api/history`	Returns chronological audit trail of all previous checks
`GET`	`/api/dashboard`	Aggregates real-time statistics computed directly from SQLite
`GET`	`/api/documents/{id}/pdf`	Generates and streams official downloadable PDF Verification Report
`POST`	`/api/compare-images`	Pixel-level difference comparison highlighting altered regions in red

7. Database Relational Schema (SQLite)

Table Name	Primary Columns & Data Types	Relational Role
`documents`	id (INT PK), document_uuid (TEXT), filename (TEXT), stored_path (TEXT), file_type (TEXT), uploaded_at (TEXT), file_size (INT), sha256 (TEXT), status (TEXT), extracted_text (TEXT), report_json (TEXT)	Stores ingested files, calculated forensic indicators, and OCR text.
`trusted_originals`	id (INT PK), original_uuid (TEXT), label (TEXT), sha256 (TEXT), filename (TEXT), category (TEXT), registered_at (TEXT)	Authoritative registry of reference fingerprints.
`verifications`	id (INT PK), document_id (INT FK), original_id (INT FK), document_name (TEXT), original_label (TEXT), result (TEXT: MATCH/MISMATCH), checked_at (TEXT), details_json (TEXT)	Historical audit trail tracking all integrity comparisons.
`audit_logs`	id (INT PK), action (TEXT), entity_type (TEXT), entity_id (TEXT), timestamp (TEXT), details (TEXT)	System-level operational traceability.

8. Experimental Validation & Alteration Detection Test

In accordance with Slide 17 of the project review, DocuShield was evaluated on synthetic document forgery pairs. The test case evaluates a genuine retail receipt against an edited copy where the transaction amount was tampered from **Rs. 2,500** to **Rs. 9,500**.

Display Amount	Rs. 2,500	Rs. 9,500	Amount text edited
SHA-256 Digest	`Distinct Master Digest`	`Different Target Digest`	**MISMATCH DETECTED**
Laplacian Sharpness	σ^2 = 1,420 (High text contrast)	σ^2 = 1,418 (Slight interpolation)	Consistent sharpness range
Canny Edge Density	3.82%	3.85%	Local glyph density shift
Visual Difference Result	Baseline reference	Modified bounding box isolated	**Altered region outlined in red**
Final System Verdict	MATCH (if registered)	Requires Review (MISMATCH)	Human review mandatory

9. Responsible Interpretation & Boundaries

A critical design philosophy of DocuShield (Slides 8, 10, and 25) is transparent, responsible interpretation:
• **Synthetic training data:** Machine learning output is experimental screening; it cannot establish legal certainty on unseen real-world document formats without verified ground truth.
• **Hash verification:** SHA-256 confirms whether two byte streams are bit-for-bit identical. It does not discern intent or semantics: re-compressing a PDF changes the hash even if text is identical.
• **Visual difference highlighting:** Pixel comparison flags genuine alterations, but can also trigger on camera rotation, scanner dust, or compression artifacts. Therefore, DocuShield presents all indicators independently as reviewer aids rather than generating an arbitrary aggregate forgery probability.

10. Conclusion & Future Roadmap

DocuShield successfully realizes an evidence-led architecture combining cryptographic exact verification with forensic heuristics, OCR text recovery, visual diff highlighting, and publication-ready audit PDF reporting. Future enhancements will incorporate QR-based asymmetric public key verification, cloud object auto-scaling (S3/PostgreSQL), and deep-learning Noiseprint extraction for generalized tamper localization.

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